Yield Analysis for EGOFET Electrode Patterning

Measuring the Impact of the PMGI/S1813 Bilayer Lift-Off Process

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Abstract

This document defines a methodology for quantifying the success rate of the EGOFET electrode patterning process. It introduces three complementary metrics: **visual yield** (microscope inspection), **electrical yield** (device functionality), and **pattern integrity** (structural quality). These metrics enable a rigorous before/after comparison between the current single-layer S1813 protocol and the proposed PMGI/S1813 bilayer protocol. A data recording template is provided for systematic tracking across fabrication batches.

Contents

1 Motivation

The primary failure mode in current EGOFET fabrication is residual gold in the Source–Drain channels after lift-off. This forces manual removal with cotton swabs, which damages electrode patterns and reduces device-to-device reproducibility.

To evaluate whether the PMGI/S1813 bilayer process improves yield, we need **quantitative metrics** that capture:

1. How many devices survive fabrication without defects.
2. How many devices are electrically functional.
3. How uniform the electrode patterns are across a batch.

These metrics also serve as a diagnostic tool: if yield drops in future batches, the data can pinpoint which fabrication step degraded.

2 Definitions

2.1 Device

A **device** is a single EGOFET structure on the Kapton substrate, consisting of:

- Source electrode (Au)
- Drain electrode (Au)
- Channel region (between Source and Drain)
- Gate electrode (Au, coplanar)
- Gate area (defined by dextran or geometry)

Each substrate typically contains multiple devices (e.g., 4–8 per chip). All devices on a substrate are counted individually.

2.2 Batch

A **batch** is a single fabrication run, identified by date and substrate number. Example: `2026-09-15_S03` = September 15, 2026, Substrate 3.

3 Yield Metrics

3.1 Metric 1: Visual Yield (Y_v)

Definition: Percentage of devices where the channel region is **completely free** of gold residue after lift-off, as determined by optical microscope inspection.

$$Y_v = \frac{N_{\text{clean channels}}}{N_{\text{total devices}}} \times 100\% \tag{1}$$

Inspection Protocol

1. Inspect each device at $\geq 20\times$ magnification after lift-off and cleaning.
2. A device is **PASS** if: no gold islands, no metal fences, no visible debris in the channel region.
3. A device is **FAIL** if: any gold residue is visible in the channel, regardless of size.
4. Record the number of PASS and FAIL devices per substrate.

3.2 Metric 2: Pattern Integrity (Y_p)

Definition: Percentage of devices where the gold electrode pattern is **structurally intact** after lift-off — i.e., no delamination, no scratching, no broken electrodes.

$$Y_p = \frac{N_{\text{intact patterns}}}{N_{\text{total devices}}} \times 100\% \tag{2}$$

Inspection Protocol

1. Inspect each device at $\geq 10\times$ magnification.
2. A device is **PASS** if: all electrodes are continuous, edges are sharp, no peeling or cracking visible.
3. A device is **FAIL** if: any electrode is broken, peeled, scratched, or has irregular edges.
4. Record the number of PASS and FAIL devices per substrate.

3.3 Metric 3: Electrical Yield (Y_e)

Definition: Percentage of devices that show **functional** transistor behavior after semiconductor deposition and gating.

$$Y_e = \frac{N_{\text{functional devices}}}{N_{\text{total devices}}} \times 100\% \tag{3}$$

Functional Criteria

A device is **functional** if all of the following are met:

1. Transfer curve (I_D vs. V_G) shows clear ON/OFF switching.
2. $I_{\text{ON}}/I_{\text{OFF}} \geq 10$.
3. Threshold voltage $|V_{\text{th}}| < 1\,\text{V}$.
4. Hysteresis between forward and reverse sweeps $\Delta V_{\text{th}} < 0.3\,\text{V}$.

3.4 Composite Yield Score

For overall process comparison, define a **composite yield**:

$$Y_{\text{composite}} = Y_v \times Y_p \times Y_e \tag{4}$$

This penalizes any failure mode. For example, if $Y_v = 95\%$, $Y_p = 98\%$, and $Y_e = 90\%$, then $Y_{\text{composite}} = 83.7\%$.

4 Experimental Design

4.1 Before/After Comparison

To demonstrate the effectiveness of the bilayer process, perform a controlled comparison:

Table 1: Experimental design for before/after comparison.

Parameter	Group A (Baseline)	Group B (Bilayer)
Photoresist	S1813 only	PMGI SF5 + S1813
Lift-off solvent	Acetone + US	DMSO at 60 °C
Number of substrates	≥ 3	≥ 3
Devices per substrate	≥ 4	≥ 4
Total devices per group	≥ 12	≥ 12
Evaporation batch	Same	Same
OSC deposition	Same	Same
Measurement protocol	Same	Same

Control Variables

To ensure a fair comparison, **all other parameters must be identical** between groups A and B:

- Same Kapton batch and cutting procedure.
- Same Si carrier wafers (or equivalent).
- Same evaporator run (if possible) or same technician with same parameters.
- Same OSC ink batch and BAMS parameters.
- Same SAM protocol (PFBT concentration, immersion time).
- Same measurement equipment and settings.

4.2 Sample Size Justification

With $n = 12$ devices per group and a baseline yield of $Y_0 = 50\%$ (estimated from current process), we can detect a ≥ 25 percentage point improvement ($Y_1 = 75\%$) with:

- Significance level $\alpha = 0.05$
- Power $1 - \beta = 0.80$

This is a conservative estimate. If baseline yield is lower (e.g., 30%), even smaller improvements become statistically significant.

5 Data Recording

5.1 Per-Device Record

For each device, record the following fields in the data template (see Appendix A):

Table 2: Per-device data fields.

Field	Type	Description
batch_id	String	Batch identifier (e.g., 2026-09-15_S03)
device_id	Integer	Device number within substrate
process	String	baseline or bilayer
date	Date	Fabrication date
visual_pass	Boolean	Channel clean after lift-off?
pattern_pass	Boolean	Electrode pattern intact?
cotton_swab	Boolean	Cotton swab used for cleanup?
notes	String	Free-text observations
Vth	Float	Threshold voltage (V)
Ion_Ioff	Float	ON/OFF current ratio
mobility	Float	Field-effect mobility (cm^2/Vs)
hysteresis	Float	ΔV_{th} (V)
functional	Boolean	Meets functional criteria?

5.2 Batch Summary

After each batch, compute:

Table 3: Batch summary fields.

Metric	Formula
Y_v	visual_pass count / total devices
Y_p	pattern_pass count / total devices
Y_e	functional count / total devices
$Y_{\mathrm{composite}}$	$Y_v \times Y_p \times Y_e$
Cotton swab rate	cotton_swab count / total devices

6 Expected Outcomes

Based on literature and the physics of the bilayer undercut:

Table 4: Expected yield improvement.

Metric	Baseline (S1813)	Bilayer (PMGI+S1813)
Y_v (visual)	40–60%	**90–100%**
Y_p (pattern)	60–80%	**95–100%**
Y_e (electrical)	30–50%	**80–95%**
Cotton swab rate	40–60%	**<5%**

Interpreting Results

- If Y_v improves but Y_e does not: the problem may be in OSC deposition or SAM, not in lift-off.
- If Y_p improves but Y_v does not: the undercut may be insufficient — increase develop time.
- If cotton swab rate drops to near zero: the bilayer is working as intended.

7 Visualization

The data should be visualized as:

1. **Bar chart**: Y_v, Y_p, Y_e side by side for baseline vs. bilayer groups.
2. **Time series**: Yield metrics across batches to track process stability.
3. **Box plots**: Distribution of V_{th}, mobility, and $I_{\mathrm{ON}}/I_{\mathrm{OFF}}$ for each group.

Example visualization code is provided in the companion Jupyter notebook (`notebooks/yield_analysis.ip`

8 References

[1] Ruiz-Molina, S. PhD Thesis, ICMAB-CSIC, 2024.

[2] Ricci, S. PhD Thesis, ICMAB-CSIC, 2020.

[3] MicroChem Corp. *PMGI Resists — Technical Data Sheet.*

[4] Fraunhofer IPMS. "OFET Substrates — Test Chips for Material Characterization." Protocol, 2026.

A Data Recording Template

The complete data template is maintained as a structured file in the `data/` directory. See `data/yield_template.md` for the field definitions and `data/raw/` for individual batch records.

A.1 Template Format

Each batch record is a YAML file named `data/raw/{batch_id}.yaml` with the following structure:

```
batch_id: "2026-09-15_S03"
date: "2026-09-15"
process: "bilayer"          # "baseline" or "bilayer"
substrate: "Kapton 75um"
photoresist: "PMGI SF5 + S1813"
lift_off: "DMSO 60C, 2.5h"
evaporation:
  Cr_thickness_nm: 5
  Au_thickness_nm: 40
  rate_Cr: 0.2              # A/s
  rate_Au: 0.5              # A/s
```

```
devices:
  - device_id: 1
    visual_pass: true
    pattern_pass: true
    cotton_swab: false
    Vth: -0.15
    Ion_Ioff: 45
    mobility: 0.8
    hysteresis: 0.12
    functional: true
    notes: ""
  - device_id: 2
    visual_pass: true
    pattern_pass: true
    cotton_swab: false
    Vth: -0.22
    Ion_Ioff: 38
    mobility: 0.7
    hysteresis: 0.18
    functional: true
    notes: "minor edge roughness"
```


A.2 Batch Summary Computation

After recording all devices, compute the batch summary:


```
batch_summary:
  total_devices: 8
  visual_pass_count: 8
  pattern_pass_count: 8
  cotton_swab_count: 0
  functional_count: 7
  Yv: 100.0               # %
  Yp: 100.0               # %
  Ye: 87.5                # %
  Ycomposite: 87.5        # %
  mean_Vth: -0.18
  std_Vth: 0.04
  mean_mobility: 0.75
  mean_Ion_Ioff: 41
```